Measuring Advice-Style Language in OLMo-2: Effects of Detection Rules and Response Length

Anonymous ACL submission

Abstract

Advice responses often contain phrases of acknowledgment, validation, professional guidance, and list organization. Their measured prevalence depends on what counts as a match and how much text a model produces. We ask: how do detection rules and response length change conclusions about advice-style language across OLMo-2 post-training? We apply a fixed lexicon–embedding detector to a 140-prompt battery spanning five conditions and public 1B and 7B checkpoints, then audit its components and output windows. A 120-sentence AI-label diagnostic finds low union-detector precision, with false positives from the embedding component; these labels are not human validation. In the recovered 700-response final-7B cache, a 64-token window reverses the advice versus emotional-control contrast for the category named "empathy-opener." Checkpoint comparisons also depend on interface and detector scope: Base-to-SFT changes confound training with prompt serialization, while same-interface SFT-to-DPO trajectories differ across marker categories. Finally, chosen-side referral/disclaimer and structuring phrase enrichment in roughly 378,000 DPO pairs per model shrinks sharply among responses within 20 words in length. The study therefore measures detector-defined language patterns and their sensitivity; it does not estimate semantic empathy, referral quality, or causal effects of training examples.

1 Introduction

Consider two answers to "I've been feeling anxious and cannot sleep—what should I do?" One begins, "That sounds really difficult. Here are a few steps," while another gives the same information without either phrase. A substring rule, a sentence-embedding rule, and a fixed 64-token window may classify these answers differently. If one condition also elicits longer answers, the detector has more sentences in which to find a match. The resulting rate is therefore a property of the responses *and* the measurement rule.

This paper asks one question: *How do detection rules and response length change conclusions about advice-style language across OLMo-2 post-training?* A **marker** is a specified textual feature, such as one of the phrases in our lexicon. A **detector** is the fixed algorithm that predicts whether a marker category occurs. A detector output is a **proxy** only when used as an indirect stand-in for a broader construct such as empathy. We use these terms consistently below. In particular, the category named *empathy-opener* in the original artifact is searched anywhere in an answer, and *referral/disclaimer* combines referrals, disclaimers, and professional-language mentions. Neither output is a semantic judgment.

OLMo-2 is useful for this measurement study because its final Instruct model is the product of four public data-bearing stages: pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2025). OLMo-2 offers an unusually complete public chain of post-training data and intermediate checkpoints. We evaluate the same fixed detector across checkpoints and use exact phrase searches in available training data as supporting context. These observations localize detector changes and co-occurrences; they do not attribute an output to a training example.

Contributions.

1. We show that conclusions about condition differences change across exact, embedding, union, and fixed-window rules; the available AI audit identifies substantial embedding-driven false positives but does not replace human labels.

2. Under the historical detector, same-interface

SFT-to-DPO changes differ by category: validation gaps increase at both model sizes, while structuring gaps contract; Base-to-SFT changes remain confounded with prompt serialization.

3. We show that aggregate DPO phrase differences are strongly associated with within-pair response length: referral/disclaimer differences approach zero and structuring differences shrink among similarly sized responses.

2 Related Work

Behavioral tests and checkpoint trajectories. Our axis-targeted prompt battery follows the behavioral-testing principle that targeted capabilities require controlled tests beyond aggregate accuracy (Ribeiro et al., 2020). Pythia makes dense public checkpoints available for training-dynamics research (Biderman et al., 2023); checkpoint studies have tracked how loss and downstream behavior change during pretraining (Xia et al., 2023). We instead compare named post-training stages and test changes in condition gaps rather than raw behavior rates.

Very recent work traces situation modeling and putative mentalizing across OLMo-2 and Pythia stages while stress-testing construct validity (Rivière et al., 2026). It is closest in model family and developmental framing; our target is different (free-form advice markers and their training-data co-occurrence), but its caution about mapping task outputs to cognitive constructs directly motivates our proxy terminology.

Empathy and emotional-support measurement. Emotional-support research typically grounds support strategies in theory and human annotation (Liu et al., 2021). Empathy research also warns that different operational definitions can make results difficult to compare (Lahnala et al., 2022). Recent studies compare human and LLM judgments directly (Welivita et al., 2026); in mental-health settings, LLM judges show reduced reliability for affective properties such as empathy (Badawi et al., 2026). These results make response-level human validation the appropriate standard. Our AI-only audit diagnoses the rule; it does not meet that standard.

Length in preference evaluation. Preference and reward-model data can encode length correlations (Singhal et al., 2023), and length-controlled evaluation can change model rankings (Dubois et al., 2024). Our contribution is narrower: we test how response length changes marker-specific exact-match differences in the OLMo-2 DPO mixtures. The post-hoc length strata describe an association and are not a causal adjustment.

Pretraining attribution and n-gram search. infini-gram (Liu et al., 2024) makes exact-substring counting over web-scale corpora tractable, and OLMoTrace (Liu et al., 2025) traces model outputs back to pretraining text. Both are pretraining-only; we add role-scoped search over the post-training layers, where our stagewise results describe transition-level changes.

Open instruction and preference data. Tulu-3 (Lambert et al., 2024) is the recipe behind OLMo-2's post-training; the SFT mixtures, DPO preference mixes, and RLVR sets are public, as is oasst1 (Köpf et al., 2023) (itself an internal subset of the Tulu-3 mixtures). Public post-training data and checkpoints make the trace auditable.

The superficial alignment hypothesis. LIMA (Zhou et al., 2023) argues alignment largely surfaces a latent style from few demonstrations; Ghosh et al. (2024) interrogate this. Our stagewise gaps give a marker-level version: each proxy is nonzero at base, and post-trained checkpoints show larger advice-minus-factual detector gaps.

Causal data attribution. LESS uses optimizer-aware gradient similarity to select influential instruction data (Xia et al., 2024); Xiao and Aranguri (2026) perform probe-based causal attribution on OLMo-2 DPO and mitigate a behavior by editing top-attributed preference pairs. We differ in kind: our trace is observational, spans multiple public post-training checkpoints and datasets, and nominates clusters and stages for such interventions; it does not establish causation.

3 Measurement and Study Design

3.1 Operational categories and fixed detector

A hand-seeded, inspectable lexicon specifies four marker categories using the artifact's historical names: *empathy-opener*, *validation-phrase*, *referral/disclaimer* (artifact key `disclaimer`), and *structuring-marker* (`structure`). The names label rules rather than validated constructs. The empathy-opener rule searches every sentence, not only an opening span. Referral/disclaimer

is a broad combined rule that also includes professional-language fragments such as "consult a" and "medical professional"; it is not a detector of referrals alone. A fifth historical category, `crisis_referral`, is excluded from the four-category analysis after producing 0 true positives and 11 false positives in the audit.

The comparisons treat the archived detector outputs and primary threshold as fixed rather than selecting rules after inspecting a contrast. The detector takes the union of case-insensitive substring matching and an e5-small embedding rule (Wang et al., 2024): a response is positive when either an exact phrase appears or any sentence reaches cosine similarity $\tau = 0.82$ to a category exemplar. We also examine $\tau = 0.80$ and 0.84. Longer responses supply more sentences and therefore more opportunities for the embedding rule to fire. The complete specification used to reproduce the July final-7B response analysis is in Appendix B; the raw-cache diagnostics preserve that version rather than retrospectively improving it. Archived stage summaries are analyzed as historical outputs. Because most stagewise raw responses are unavailable, we do not claim to have re-executed every stage under a byte-identical detector.

Construction and pilot overlap. The released pilot file contains eight advice prompts and four factual controls used while hand-seeding the proxy; all 12 reappear in the 140-prompt battery. We therefore do not call the full battery held out. A final-7B sensitivity excluding those 12 leaves the advice–factual gaps similar (Appendix D). The sample sizes refer to distinct datasets: the 140 prompts form the marker evaluation, whereas 120 stratified final-model sentences form a detector-accuracy audit. We audit those sentences (40/30/20/20/10 across the five conditions) using two separately run, blinded Codex passes and Claude Opus 4.8 adjudication of their three disagreements. The artifact records the Opus identifier and run metadata; exact Codex model/build identifiers were not retained. This is an AI diagnostic, not human validation. The labels, adjudication reasons, and detector are released (Section 4.1; Appendix B).

3.2 Prompt conditions and generation

Five conditions target different aspects of advice-style responses: *advice* (60 personal or emotional help requests), *factual* (30 knowledge questions), *emotional non-advice* (20 descriptive prompts about emotional topics without a help request), *neutral advice* (20 how-to requests without emotional stakes), and *domain factual* (10 impersonal clinical-information questions). These controls target conceptual axes but are not paired counterfactuals or topic-matched scenarios; non-significance is not interpreted as equivalence. The final-model battery and the canonical advice/factual checkpoint trajectories use five sampled generations per prompt (temperature 0.7, top-p 0.9, 256 new tokens, distinct seeds). The Base/SFT/DPO axis-targeted extension uses four; its balanced comparisons use seeds 0–3 for every condition.

3.3 Inference at the prompt level

The clustering unit is the prompt: each prompt contributes its mean detector-positive indicator over its four or five generations, and condition rates are means over prompts. All between-condition inference uses prompt-clustered permutation tests (10,000 permutations of prompt-level means; two-sided; Monte Carlo $p = (b + 1)/(B + 1)$). Benjamini–Hochberg correction is applied separately to the 24 final-condition contrasts, 28 advice/factual adjacent-stage interactions, 24 axis-targeted stage interactions, and 8 primary DPO pairwise tests. We do *not* treat generations as independent Bernoulli trials: all condition and stage tests operate on prompt-level generation means.

3.4 Checkpoint evaluation and supporting corpus search

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct; 7B Base→SFT→DPO→Instruct. Base checkpoints have no chat template and use a plain Question/Answer format, whereas post-trained checkpoints use their official chat template. Thus Base→SFT is a stage-plus-interface transition and cannot isolate SFT training. No serialization crossover was run; we demote that transition throughout. The stagewise grid runs the 90 advice+factual prompts at every checkpoint. All five conditions run on the final models and, in the four-seed specificity extension, at Base, SFT, and DPO.

Corpus searches play a supporting role. For each stage's available indexed data we search the exact Stage-1 corpus (`v4_olmo-mix-1124_llama`; novelty = 1−verbatim coverage), each model's exact SFT mixture (`tulu-3-sft-olmo-2-mixture-0225`

for 1B; `tulu-3-sft-olmo-2-mixture` for 7B) in DuckDB with assistant-only scoping (866,138 conversations for 1B and 939,344 for 7B; oasst1 is not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares seeded marker phrases against length-matched topical controls from the same answers, per-million documents, with a distinct-phrase leave-one-out. Stage-2 mid-training uses Dolmino-Mix-1124, for which no public index was available; the corpus trace is therefore incomplete.

Reproducibility. Stagewise, condition, and corpus numbers derive from `summary_v3.json` and released scripts under `experiments/`; prompts, lexicon, seeds, thresholds, and adjusted-p output ship with it. Audit numbers are separately materialized in `ai_consensus_validation_v3.json`, with both 120-item passes, consensus labels, Opus decisions, and reasons released alongside it.

4 Results

We first audit the measurement rule, then report condition and checkpoint comparisons under that rule, and finally examine the association between DPO phrase differences and response length. This order matters because the audit limits what the subsequent detector-positive rates can mean.

4.1 Measurement audit: precision, components, and output windows

A condition-stratified random audit samples 120 final-model sentences across all five conditions (40 advice, 30 factual, 20 emotional non-advice, 20 neutral advice, and 10 domain factual). The sample contains 56 sentences from the 1B model and 64 from the 7B model, drawn from 97 distinct seed-0 responses. It was not stratified on detector predictions, and multiple sentences can come from one response. Two independent blinded Codex passes agree exactly on 117 of 120 sentences (97.5%). Category agreement is 99.2–100%; Cohen's κ is .885 for empathy expression, .663 for validation, 1.000 for referral, and .978 for structuring. Claude Opus 4.8 adjudicates the three disagreements. These are AI-consensus labels under specific prompts and models, not human ground truth.

Against that consensus, the union detector has low precision and category-dependent recall (Table 1). The audit's referral label is narrower than the detector: it requires an actual recommendation to seek a qualified professional, while the historical referral/disclaimer rule also matches disclaimers and professional-language mentions. Its 0.06 precision therefore reflects both broad detector behavior and target misalignment; it should not be read as the accuracy of a pure referral detector. Empathy and validation recall are each 0.50. Errors vary by condition, so subtracting condition rates does not guarantee cancellation. The detector overpredicts positives in this sample, although the rates are not formal upper bounds because recall is below one.

Exact substring matching has no false positives in these 120 sentences but misses many AI-labeled positives. Its true-positive/false-positive/false-negative counts are 2/0/2 for empathy expression, 1/0/1 for validation, 1/0/0 for referral, and 9/0/21 for structuring. Every union false positive is introduced by the embedding component. The union raises structuring recall from 0.30 to 0.87, with lower precision. Component choice is therefore part of the measurement question.

The recovered 700-response cache permits a separate response-level diagnostic for the final 7B checkpoint only. Advice responses average 244 tokens and factual responses 187; the three targeted-control means are 251–255 tokens. Restricting each response to its first 64 tokens changes the union detector's advice-minus-emotional contrast for the category historically called empathy-opener from $-.147$ to $+.253$. Validation remains positive, moving from .250 to .177; referral/disclaimer moves from .090 to .113; and structuring remains near zero, moving from $-.030$ to $-.033$. On full responses, exact-only advice-minus-targeted-control contrasts are .323 for empathy, .153 for validation, .157 for referral/disclaimer, and .390 for structuring. These component and window checks qualify the final-7B result; raw responses are unavailable for the other checkpoints, so they do not validate the stagewise trajectories.

4.2 Condition comparisons depend on the measurement rule

Table 2 gives mean prompt-level proxy-positive rates per condition on the final models; Table 3 tests advice against each axis-targeted control. Against plain factual controls every category separates sharply in both models (all prompt-clustered $p \leq$

category	P	R	F1	tp/fp/fn
empathy expr.	0.22	0.50	0.31	2/7/2
validation	0.33	0.50	0.40	1/2/1
ref./discl.	0.06	1.00	0.12	1/15/0
structuring	0.49	0.87	0.63	26/27/4

Table 1: Union-detector precision, recall, and F1 against the adjudicated 120-sentence AI consensus ($\tau = 0.82$). This is a diagnostic against an AI annotation construct, not human validation. The referral annotation is narrower than the combined historical referral/disclaimer rule. Response-cluster bootstrap intervals appear in Appendix H.

.0002). For referral/disclaimer, the 1B model has rates .900 on advice and .080 on factual prompts; the 7B model has rates .913 and .060.

Targeted controls change the interpretation. At $\tau = .82$, the category historically called empathy-opener fires more often on emotional non-advice than advice: .790 versus .590 for 1B ($p = .010$), and .760 versus .617 for 7B ($p = .098$). As the preceding audit shows, the final-7B direction reverses in a 64-token window. Structuring is high on both advice forms: the personal-minus- neutral advice difference is .000 for 1B ($p = 1.0$) and $-.030$ for 7B ($p = .574$). These are tests for a difference, not equivalence tests.

The broad referral/disclaimer category is also common outside personal advice. It fires on .740 of neutral-advice prompts for 1B and .720 for 7B, and on .820 of domain-factual prompts for both models. Its advice-minus-domain-factual difference is small and not significant: .080 for 1B ($p = .352$) and .093 for 7B ($p = .159$). The validation-phrase union output is higher on advice than every targeted control for both models (all unadjusted $p \leq .0044$; all BH-adjusted $q \leq .0082$). These marginal profiles differ, but weak correlations do not establish the absence of a latent response template. The comparisons are detector-dependent, and the small, mixed 1B–7B differences do not support a scaling claim.

Marker co-occurrence. The recovered 700-response final-7B cache permits response-level co-occurrence analysis at $\tau = .82$. Among advice responses, 4.7% contain one proxy, 29.0% two, 40.7% three, and 25.7% all four (none contain zero). Referral/disclaimer is present in 93.0% of validation-positive advice responses, while pairwise phi correlations range from $-.14$ to .21. The categories often co-occur, but these weak correlations alone neither establish nor rule out a latent response template. Condition-level joint distributions and all pairwise correlations are released (Appendix D).

4.3 Checkpoint contrasts under the historical detector

Raw advice-side rates rising across stages would be ambiguous because post-training could simply make *all* outputs more cautious and listy. A more informative operational quantity is the advice-minus-factual gap Δ_s at each stage (Table 5) and its *change* across adjacent stages, a stage$\times$condition difference-in-differences tested with a prompt-clustered permutation test on per-prompt stage trajectories (Table 4; 10k permutations, with 60 advice prompts and 30 factual prompts).

All eight Base-to-SFT/interface changes are positive and six survive BH correction. For referral/disclaimer, the change is +.253 for the 1B model and +.243 for the 7B model; for structuring it is +.337 and +.227, respectively. Validation does not survive correction ($q = .137$ for 1B and $q = .121$ for 7B). Every category is already detector-positive at Base. Because Base uses plain QA and SFT uses the chat template, these changes cannot isolate training from serialization.

The same-interface SFT-to-DPO transition differs by category. The validation gap increases by +.213 for 1B ($q = .002$) and +.147 for 7B ($q = .021$). The 1B referral/disclaimer gap increases by +.190 ($q = .002$), while its 7B change of +.083 is not significant after correction ($q = .203$). The empathy-opener changes are also not significant after correction. The structuring gap contracts by $-.367$ for 1B and $-.277$ for 7B (both $q = .002$), because DPO raises detector-positive structuring on factual answers while advice-side rates are near ceiling. Aggregate preference-data co-occurrence cannot independently explain these trajectories because its referral difference approaches zero among similar-length pairs (Table 6).

No RLVR or final-merge gap change is significant (all raw $p \geq .168$). These tests describe the outputs of the fixed detector. Because the audit finds condition-dependent false positives and does not span checkpoints, they do not by themselves establish changes in human-judged behavior prevalence.

Comparisons with targeted controls. The four-seed Base/SFT/DPO extension tests 24 planned

Marker proxy	Model	Advice	Factual	Emotional	Neutral adv.	Domain fact.
empathy opener	1B	0.590	0.040	**0.790**	0.020	0.340
	7B	0.617	0.027	**0.760**	0.070	0.360
validation phr.	1B	0.363	0.007	0.120	0.000	0.040
	7B	0.383	0.013	0.130	0.030	0.060
ref./discl.	1B	0.900	0.080	0.510	0.740	0.820
	7B	0.913	0.060	0.540	0.720	0.820
structuring	1B	1.000	0.580	0.820	**1.000**	0.920
	7B	0.970	0.547	0.830	**1.000**	0.840

Table 2: Mean prompt-level union-proxy rate per condition (final models, $\tau = 0.82$; each prompt averaged over 5 generations). Bold marks the cells that break the "advice-specific" reading: the empathy-opener proxy peaks on emotional *non-advice* prompts, and structuring is at ceiling on *neutral* advice. The audit sample shows substantial overprediction, suggesting upward-biased absolute rates (Section 4.1); `crisis_referral` is excluded (0 TP / 11 FP in the 120-sentence audit).

Marker	Model	−Emot.	−Neut.adv.	−Dom.fact.
empathy	1B	−.200 (.010)	.570 (.0002)	.250 (.016)
	7B	−.143 (.098)	.547 (.0002)	.257 (.031)
validation	1B	.243 (.003)	.363 (.0002)	.323 (.003)
	7B	.253 (.003)	.353 (.0003)	.323 (.004)
referral	1B	.390 (.0002)	.160 (.006)	.080 (.352)
	7B	.373 (.0002)	.193 (.001)	.093 (.159)
structuring	1B	.180 (.0002)	.000 (1.0)	.080 (.018)
	7B	.140 (.004)	−.030 (.574)	.130 (.026)

Table 3: Advice minus each axis-targeted control (prompt-clustered permutation p, 10k draws). Benjamini–Hochberg correction across the 24 displayed tests leaves the validation-phrase result unchanged (all $q \leq .0082$). Non-significant cells mean "no detected difference," not equivalence.

Model	Marker proxy	Base→SFT/interface	SFT→DPO
1B	empathy opener	+0.190 ($q = .004$)	+0.117 ($q = .121$)
1B	validation phrase	+0.093 ($q = .137$)	+0.213 ($q = .002$)
1B	referral/disclaimer	+0.253 ($q = .002$)	+0.190 ($q = .002$)
1B	structuring	+0.337 ($q = .002$)	−0.367 ($q = .002$)
7B	empathy opener	+0.263 ($q = .002$)	+0.070 ($q = .387$)
7B	validation phrase	+0.103 ($q = .121$)	+0.147 ($q = .021$)
7B	referral/disclaimer	+0.243 ($q = .002$)	+0.083 ($q = .203$)
7B	structuring	+0.227 ($q = .002$)	−0.277 ($q = .002$)

Table 4: Change in the advice-minus-factual proxy gap across transitions (plus-one-corrected prompt-clustered permutation tests, 10k draws; displayed q is BH-adjusted across all 28 adjacent-stage×proxy tests). The Base→SFT column also changes the input serialization and is not a clean SFT effect. Later raw $p \geq .168$.

stage×control interactions (Appendix E). Four survive BH correction at $\tau = .82$. For empathy relative to emotional non-advice, the 7B Base→SFT/interface transition increases the advice gap ($+0.358$, $q = .016$), but DPO reverses it at both 1B (-0.467, $q = .001$) and 7B (-0.642, $q = .001$). At 7B, SFT also shifts structuring toward neutral advice (-0.208, $q = .026$). No validation or referral interaction survives $q < .05$; 1B DPO validation tendencies have $q = .059$–$.095$. The threshold sweep yields three significant interactions at $\tau = .80$, four at .82, and five at .84; only the two 7B empathy transitions survive at all three thresholds. Thus the broad advice-minus-factual changes cannot all be read as advice-specific.

4.4 DPO phrase differences track response length

The exact mixtures contain 378,301 preference pairs for the 1B model and 378,341 for the 7B model. Chosen responses are slightly shorter in aggregate: the chosen-to-rejected mean-word ratio is .990 for 1B and .997 for 7B. This aggregate normalization does not remove within-pair length as a possible covariate. At pair level, referral/disclaimer matches have absolute chosen-minus-rejected effects of 0.100 percentage points (pp; 95% CI $[.065, .135]$) at 1B and 0.179 pp ($[.144, .214]$) at 7B; matched odds ratios (MORs) are 1.181 and 1.352. Structuring effects are larger: 1.393 pp ($[1.278, 1.507]$) and 2.280 pp ($[2.167, 2.394]$). Empathy is weakly enriched only at 1B after correction; validation is not (Table 6).

Only 19.1% of 1B pairs and 18.7% of 7B pairs fall within ±20 words. In that comparable-length stratum, the referral/disclaimer difference is $-.010$ pp for 1B (MOR .97, raw $p = .763$) and +.009 pp for 7B (MOR 1.03, raw $p = .810$). The high-precision referral difference is +.017 pp for 1B ($p = .299$) and $-.003$ pp for 7B ($p = .925$). Structuring remains positive but smaller: +.184 pp for 1B (raw $p = .047$) and +.343 pp for 7B (raw $p = .0002$). In the noncomparable strata, signs follow the longer response side. Aggregate DPO enrichment is therefore strongly length-aligned;

Model	Stage	Empathy	Validation	Referral	Structuring
1B	base	.217 (.204)	.073 (.073)	.483 (.430)	.757 (.430)
1B	sft	.393 (.393)	.167 (.167)	.697 (.684)	.847 (.767)
1B	dpo	.550 (.510)	.380 (.380)	.933 (.873)	1.000 (.400)
1B	rlvr	.567 (.507)	.410 (.410)	.923 (.836)	1.000 (.440)
1B	instruct	.590 (.550)	.363 (.356)	.900 (.820)	1.000 (.420)
7B	base	.230 (.230)	.087 (.080)	.577 (.524)	.853 (.486)
7B	sft	.493 (.493)	.183 (.183)	.767 (.767)	.787 (.714)
7B	dpo	.597 (.564)	.330 (.330)	.923 (.850)	.977 (.437)
7B	instruct	.617 (.590)	.383 (.370)	.913 (.853)	.970 (.423)

Table 5: Stagewise advice detector-positive rate with the **advice-minus-factual proxy gap in parentheses** ($\tau = 0.82$). Transition tests and corrected values appear in Table 4. Base checkpoints use plain QA; all later checkpoints use chat serialization, so Base→SFT differences conflate stage and interface.

Model	Marker proxy	Δ pp	MOR	Chosen only	Rejected only	BH q
1B	empathy opener	.035	1.082	1768	1634	.036
1B	validation phrase	.006	1.130	183	162	.282
1B	referral/disclaimer	.100	1.181	2478	2098	$< 10^{-7}$
1B	structuring	1.393	1.242	27047	21779	$< 10^{-124}$
7B	empathy opener	.030	1.069	1734	1622	.074
7B	validation phrase	.009	1.224	191	156	.078
7B	referral/disclaimer	.179	1.352	2602	1925	$< 10^{-22}$
7B	structuring	2.280	1.435	28466	19838	$< 10^{-300}$

Table 6: Aggregate DPO pair-level effects, unadjusted for within-pair length. Δ pp is the absolute paired risk difference in percentage points; MOR is chosen-only/rejected-only. q is Benjamini–Hochberg adjusted across eight exact McNemar tests. Confidence intervals and descriptive per-word ratios are reported in the text and artifact.

marginal per-word ratios do not establish a within-pair preference effect. Conditioning on length also changes the target quantity and does not identify a causal preference for a marker.

A narrower referral lexicon that excludes the broad "consult/talk/speak to a" phrases remains positive: 0.023 pp ($[.006, .040]$, MOR 1.17, raw $p = .010$) at 1B and 0.043 pp ($[.026, .061]$, MOR 1.34, $p < 2 \times 10^{-6}$) at 7B. The original referral direction is positive in eight of nine 1B sources and all nine 7B sources; the high-precision direction is positive in six of nine 1B sources and seven of nine 7B sources. Structuring is positive in eight of nine sources at each size and is driven mainly by "first," and "here are some." Only 5.7% of rows belong to a duplicated prompt-text group (maximum multiplicity 15), so ordinary row-level McNemar inference may be mildly optimistic but is not dominated by repeated prompts. Full phrase/source results and intervals are in Appendix G. These co-occurrences are compatible with, but do not attribute or directly validate, checkpoint changes.

Supporting SFT and RLVR searches. The assistant-only SFT searches recover seeded phrases, but the comparison is phrase-sparse and dominated by "here are some"; per-phrase and leave-one-out results are in Appendix F. The searched RLVR datasets contain zero or one match for each referral, empathy, and validation phrase, while generic structuring markers remain. These exact searches establish phrase co-occurrence in available data, not attribution or a training effect.

5 Discussion

What the trajectories measure. The checkpoint analysis reveals trajectories of a fixed historical detector, not semantic advice behaviors. All advice-minus-factual gaps increase across the confounded Base-to-SFT/interface transition, and same-interface SFT-to-DPO changes vary across categories. The detector audit shows why the distinction matters: embedding matches add false positives, and a fixed output window can reverse a key final-7B contrast. The available evidence therefore supports comparisons among explicitly named rules and windows, while human-judged checkpoint trajectories remain open.

Length and preference data. Exact matching avoids embedding-threshold errors in the DPO analysis, but it still gives longer responses more opportunities to contain a phrase. Aggregate referral/disclaimer and structuring differences largely follow the longer side of a pair. Among responses within 20 words, referral/disclaimer differences are near zero and structuring differences are much smaller. Because the length strata are post hoc and conditioning on length changes the target quantity, this analysis diagnoses sensitivity rather than a causal preference for or against any marker.

Implications for measurement. Broad historical category names should not be mistaken for narrow constructs. The referral/disclaimer rule combines recommendations, disclaimers, and professional-language mentions; the empathy-opener rule searches beyond the opening sentence. Corpus matches show where the same strings occur, but do not identify the examples responsible for a model output. Future work should align category definitions with independent human labels, annotate response prefixes when evaluating prefix detectors, and cross checkpoint with prompt interface before making semantic or training-stage claims.

6 Conclusion

Conclusions about advice-style language in OLMo-2 change with the detector and the amount of response text examined. The historical union detector produces distinct condition and checkpoint trajectories, but its low precision, interface confounding, and final-7B window sensitivity preclude a semantic reading. Exact DPO matches are more common on the chosen side in aggregate; among responses within 20 words, referral/disclaimer differences approach zero and structuring differences shrink from 1.393–2.280 percentage points to .184–.343. These results support a focused measurement claim: detection rules and response length materially alter the observed pattern. They do not show semantic empathy prevalence, referral quality, or causal effects of training.

7 Limitations

All analyses are observational. Checkpoint localization, corpus co-occurrence, and DPO chosen/rejected differences establish neither training-example attribution nor a causal effect. Base uses plain QA while SFT and later checkpoints use chat serialization, so every Base-to-SFT contrast combines stage and interface.

The 120-sentence audit uses AI-consensus labels, has no completed human pass, and does not validate response-level prevalence. It samples 97 seed-0 responses, and category-level intervals are wide. Raw-response window checks are available only for the final 7B checkpoint. They cannot repair or validate trajectories at the other checkpoints.

The targeted controls are not topic-paired to the 60 advice prompts, and a non-significant contrast is not evidence of equivalence. Twelve pilot prompts were reused in the 140-prompt battery; excluding them changes final-7B advice-minus-factual gaps by at most .045. Five canonical seeds and four extension seeds do not remove sampling uncertainty.

The DPO length bins are post hoc rather than a pre-specified conditional model, and 5.7% of rows belong to duplicated prompt-text groups. SFT recoverability is based on only three to six distinct three-word types. Stage-2 Dolmino-Mix-1124 was not searched because no public index was available, and the Stage-1 novelty measure is verbatim-only and length-confounded. We study one model family, two sizes, one decoding configuration, and modest author-written prompt sets; no scaling or cross-family claim follows.

Ethics Statement

Data and human subjects. This work uses only publicly released artifacts (OLMo-Mix-1124 via the public infini-gram API; the OLMo-2 Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets; open-weight OLMo-2 models and their public intermediate checkpoints). We collected no new data from people. All 140 prompts are short, author-written strings describing hypothetical situations with no personal information. The study involves no human subjects and needed no ethics-board review.

Sensitive domain. Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns surface phrases that can mark safety-relevant moves such as professional referrals. The audit sample shows substantial detector overprediction, suggesting upward-biased absolute rates; we also drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

Dual use. Making the provenance of a safety-relevant marker inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and release an analysis tool, not any model or dataset that could cause direct harm.

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A Prompts and decoding configuration

All answers use nucleus sampling (temperature 0.7, top-p 0.9, `max_new_tokens=256`, `do_sample=True`), the chat template where available (plain `Question:/Answer:` for base checkpoints), and five seeds per prompt in the canonical grid. The battery: 60 advice, 30 factual, 20 emotional non-advice, 20 neutral advice, 10 domain factual; the complete machine-readable list ships as `experiments/prompts_full.json`.
The Base/SFT/DPO axis-targeted-control extension uses four seeds; all balanced extension tests use seeds 0–3 for every condition. Eight advice and four factual prompts were reused from the earlier 12-prompt pilot; the other 128 are new to the full battery.

Examples.

- Advice: `I've been feeling really anxious lately and can't sleep. Any advice?`
- Factual: `How does photosynthesis work?`
- Emotional non-advice: `Describe the common symptoms of loneliness.`
- Neutral advice: `How should I organize my bookshelf?`
- Domain factual: `What are common clinical treatments for insomnia?`

B Surface-marker lexicon and detector threshold

This section records the executable four-category detector in `experiments/raw_output_robustness.py`, used to reproduce the July final-7B response analysis. Historical notebook variants also included a crisis category, which is excluded from the reported four-category analysis after its audit failure. A category fires for an answer if *either* (i) any lexicon phrase occurs as a case-insensitive substring, *or* (ii) any sentence has cosine similarity $\geq \tau$ to a seed exemplar (`intfloat/multilingual-e5-small`, `query:`/`passage:` prefixes), with $\tau = 0.82$ primary. Every sentence is eligible; despite its historical name, `empathy_opener` is not restricted to an opening span.

Lexicon (substring match).

- **empathy-opener proxy** (`empathy_opener`): `i'm sorry to hear`, `i am sorry to hear`, `i'm really sorry`, `that sounds really`, `that sounds very`, `i can understand`, `i can imagine`, `it's understandable`, `that must be`, `i'm here for you`, `i hear you`, `it sounds like you`
- **validation-phrase proxy**: `your feelings are valid`, `it's okay to feel`, `it is okay to feel`, `it's normal to feel`, `it's completely normal`, `what you're feeling`, `there's nothing wrong with`, `it's natural to`
- **referral/disclaimer proxy** (`disclaimer`): `consult a`, `talk to a`, `speak with a`, `see a doctor`, `seek medical`, `i am not a doctor`, `i'm not a doctor`, `i'm not a medical`, `medical professional`, `healthcare professional`, `mental health professional`, `seek professional`, `professional help`
- **structuring-marker proxy** (`structure`): `here are some`, `here are a few`, `consider the following`, `steps you can take`, `a few things you can`, `first,`, `secondly,`

e5 seed exemplars. Empathy: "I'm so sorry you're going through this"; "That sounds incredibly hard"; "I can only imagine how painful this is." Validation: "Your feelings are completely valid"; "It's totally normal to feel this way"; "There's nothing wrong with feeling upset." Referral/disclaimer: "You should consult a qualified professional"; "Please talk to your doctor about this"; "I'm not a medical professional, but." Structuring: "Here are a few steps you can take"; "Consider the following suggestions"; "First, try this; second, try that."

C Threshold sweep

Table 7 reports the exact contrasts at $\tau = 0.80$, 0.82, and 0.84. Structuring is stable: advice and neutral-advice remain within .05 in every cell. Validation exceeds its strongest control in all six cells. Empathy's emotional-minus-advice contrast reverses at .84, while referral's advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was fixed before the 140-prompt battery, but its qualitative pattern should not be generalized across detector thresholds.

Model	Contrast	.80	.82	.84
1B	empathy: emot.−adv.	.030	.200	−.107
7B		.030	.143	−.280
1B	validation: adv.−max ctrl.	.030	.243	.107
7B		.047	.253	.170
1B	referral: adv.−domain	−.003	.080	.093
7B		.017	.093	.280
1B	structuring: adv.−neutral	.000	.000	.047
7B		.000	−.030	−.010

Table 7: Threshold robustness of the four decomposition contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on advice and neutral advice.

D Final-7B response-level robustness

The recovered `final_7b_answers.jsonl` contains all 700 final-7B responses (140 prompts $\times$ five seeds). Table 8 shows the opportunity imbalance. "Cap proxy" means at least 255 tokens under the released OLMo tokenizer; it indicates likely truncation but is not a direct generation-stop record.

Table 9 recomputes each targeted-control contrast using exact substring matching alone and the exact-or-embedding union, over full answers and fixed prefix windows. The full-union column exactly reproduces the fixed final-7B result. The first-64-token empathy reversal shows that its full contrast is not opportunity-robust. Exact-only matching instead makes structuring look advice-specific relative to neutral advice; the embedding component removes that contrast.

Condition	n	mean tok.	mean sent.	cap proxy
advice	300	243.8	13.5	.880
factual	150	187.4	7.8	.573
emotional	100	253.5	10.9	.980
neutral advice	100	254.6	13.8	.970
domain factual	50	250.5	10.2	.960

Table 8: Final-7B response length and generation-cap proxy.

Proxy contrast	Exact full	Union full	Union 64	Union 128
empathy: adv.−emot.	.323	−.147	.253	.063
validation: adv.−emot.	.153	.250	.177	.210
referral: adv.−domain	.157	.090	.113	.117
structuring: adv.−neutral	.390	−.030	−.033	−.030

Table 9: Detector-component and fixed-window sensitivity at final 7B ($\tau = .82$ for union columns).

Among advice responses, union detections include one category in 4.7% of cases, two in 29.0%, three in 40.7%, and four in 25.7%. Pairwise phi correlations range from $-.14$ to .21; referral/disclaimer appears in 93.0% of validation-positive responses. Excluding the eight advice and four factual pilot prompts changes the empathy advice-minus-factual gap from .587 to .542, validation from .367 to .369, referral/disclaimer from .850 to .831, and structuring from .423 to .458.

E Checkpoint interactions with targeted controls

Table 10 reports every planned interaction in the four-seed Base/SFT/DPO extension. Columns Base, SFT, and DPO are advice-minus-control gaps; transition columns are changes in those gaps with BH-adjusted q in parentheses. The family contains all 24 displayed transition tests.

F Per-phrase recoverability and leave-one-out

Per-phrase assistant-only counts against each model's exact mixture (`-0225`, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl` = blob/assistant.

Length-matched buckets (occurrence-weighted). Seeded-marker vs topical per-million for 1B: 2-word 1884.3 vs 1760.2 (1.1×, n_{occ}=14), 3-word

Model	Marker	Control	Base	SFT	DPO	Base→SFT	SFT→DPO
1B	referral	domain factual	.188	.308	.167	+.121 (.414)	−.142 (.296)
1B	empathy	emotional	.017	.225	−.242	+.208 (.095)	−.467 (.001)
1B	structuring	neutral advice	−.013	−.050	.000	−.037 (.653)	+.050 (.475)
1B	validation	domain factual	.071	.146	.375	+.075 (.475)	+.229 (.095)
1B	validation	emotional	.021	.121	.225	+.100 (.293)	+.104 (.296)
1B	validation	neutral advice	.058	.171	.350	+.113 (.189)	+.179 (.059)
7B	referral	domain factual	.321	.067	.200	−.254 (.166)	+.133 (.293)
7B	empathy	emotional	.108	.467	−.175	+.358 (.016)	−.642 (.001)
7B	structuring	neutral advice	.050	−.158	−.025	−.208 (.026)	+.133 (.095)
7B	validation	domain factual	.092	.175	.317	+.083 (.475)	+.142 (.296)
7B	validation	emotional	.054	.125	.179	+.071 (.414)	+.054 (.537)
7B	validation	neutral advice	.092	.163	.292	+.071 (.414)	+.129 (.189)

Table 10: Complete targeted-control checkpoint interactions at $\tau = .82$ (10k prompt-clustered permutations). Four of 24 transition tests have $q < .05$. Only the 7B empathy Base→SFT and SFT→DPO effects retain $q < .05$ at all three detector thresholds.

Model	Phrase	Asst.	Blob	Infl.
1B	first,	44422	52676	1.19
1B	here are some	11914	13276	1.11
1B	here are a few	5421	6056	1.12
1B	healthcare professional	3075	3216	1.05
1B	mental health professional	2109	2186	1.04
1B	professional help	1765	1825	1.03
1B	seek professional	1353	1377	1.02
1B	it sounds like you	1130	1164	1.03
1B	consult a	891	978	1.10
1B	it's understandable	218	227	1.04
1B	it's natural to	152	157	1.03
7B	here are some	12049	13432	1.11
7B	here are a few	5511	6156	1.12
7B	healthcare professional	3106	3309	1.07
7B	i'm really sorry	2369	2378	1.00
7B	mental health professional	2120	2214	1.04
7B	professional help	1774	1838	1.04
7B	seek professional	1355	1379	1.02
7B	it sounds like you	1139	1184	1.04
7B	consult a	927	1031	1.11
7B	talk to a	286	408	1.43
7B	it's completely normal	78	103	1.32

Table 11: Per-phrase SFT-assistant recoverability (representative rows). The 3 (1B) / 6 (7B) distinct 3-word types drive Section 4.4; "here are some" dominates the 3-word mean.

12900.2 vs 629.8 (20.5×, 45), 4-word 1596.9 vs 11.7 (136.9×, 30). For 7B, 2-word 1784.5 vs 996.8 (1.8×, 20), 3-word 9354.4 vs 1280.4 (7.3×, 55), 4-word 1396.6 vs 96.2 (14.5×, 24). **Distinct-phrase leave-one-out (3-word):** dropping "here are some" moves the mean from 5455.2 to 1305.2/M at 1B (3 types) and from 3026.0 to 1065.9/M at 7B (6 types).

G DPO phrase and source robustness

Table 12 gives paired risk-difference and matched-odds-ratio intervals. The higher-precision referral set requires an explicit doctor or professional object (e.g., "talk to your doctor," "consult a healthcare professional," "seek professional help") and excludes broad fragments such as "consult a" and "talk to a." Intervals use the discordant-pair normal approximation for the risk difference and log-MOR; exact McNemar p-values are released.

Table 13 is the requested within-pair length diagnostic. The near-antisymmetric signs in the longer-response strata show that the aggregate chosen-side effect is strongly associated with relative response length. The comparable-length band is descriptive and post hoc; it is not a substitute for a pre-specified conditional model.

For the original referral group, positive chosen-minus-rejected differences occur in eight of nine 1B sources (one tie) and all nine 7B sources. Under the higher-precision group they occur in six of nine 1B sources and seven of nine 7B sources. Structuring is positive in eight of nine sources at each size; the negative source is the instruction-following personas subset. The main structuring contribution comes from "first," (.931 pp for 1B and 1.381 pp for 7B) and "here are some" (.339 pp for 1B and .736 pp for 7B), rather than every phrase moving uniformly. The main referral contributors are "professional help," "consult a," and named professional phrases. There are 367,082 distinct 1B prompt texts and 367,110 distinct 7B prompt texts; 21,551 1B rows and 21,574 7B rows (5.7% for each) belong to duplicated groups, with maximum multiplicity 15.

H Response-cluster audit uncertainty

The 120 audit sentences come from 97 seed-0 responses. To preserve dependence among sentences from the same response, we assign Dirichlet(1) weights to response keys in a 10,000-

Model	Phrase group	Δ pp [95% CI]	MOR [95% CI]	C-only	R-only
1B	referral/disclaimer	.100 [.065,.135]	1.181 [1.114,1.252]	2478	2098
1B	high-precision referral	.023 [.006,.040]	1.170 [1.039,1.318]	591	505
1B	structuring	1.393 [1.278,1.507]	1.242 [1.220,1.264]	27047	21779
7B	referral/disclaimer	.179 [.144,.214]	1.352 [1.274,1.434]	2602	1925
7B	high-precision referral	.043 [.026,.061]	1.335 [1.187,1.502]	649	486
7B	structuring	2.280 [2.167,2.394]	1.435 [1.409,1.461]	28466	19838

Table 12: DPO paired absolute and relative effects. Positive Δ means more chosen-only than rejected-only matches.

Model	Word-difference band	n	Referral		High precision		Structuring	
			Δ pp	MOR	Δ pp	MOR	Δ pp	MOR
1B	chosen longer > 20	171170	+.470	1.991	+.059	1.408	+5.979	2.361
1B	rejected longer > 20	134707	$-.310$	.616	$-.020$	.871	-3.786	.579
1B	within ± 20	72424	$-.010$	.965	+.017	1.238	+.184	1.062
7B	chosen longer > 20	187201	+.502	2.172	+.098	1.761	+6.562	2.610
7B	rejected longer > 20	120381	$-.223$	.706	$-.015$	.905	-3.239	.619
7B	within ± 20	70759	+.009	1.028	$-.003$	.966	+.343	1.123

Table 13: DPO effects stratified by chosen-minus-rejected response length. Positive Δ means more chosen-only matches.

Category	Precision 95%	Recall 95%	F1 95%
empathy expression	[.04,.46]	[.10,.90]	[.05,.57]
validation	[.02,.84]	[.03,.98]	[.02,.83]
referral	[.00,.22]	[1.00,1.00]	[.00,.36]
structuring	[.35,.63]	[.74,.96]	[.49,.74]

Table 14: Bayesian response-cluster bootstrap intervals for the 120-sentence AI consensus audit. Referral recall is degenerate because the single positive gold item is detected in every draw; its precision and F1 remain highly uncertain.

draw Bayesian bootstrap and recompute each metric. Table 14 reports percentile intervals. These intervals quantify sampling uncertainty around the AI consensus labels; they do not turn the audit into human validation.

I Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b + 1)/(B + 1)$; $p < 10^{-4}$ denotes zero raw exceedances. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean seeded-marker per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs, Haldane–Anscombe matched odds ratios with log-scale intervals, paired risk differences with normal-approximation intervals, and phrase occurrences per million whitespace-delimited response words. Released scripts derive all statistics from the artifact.